

On the Traces of Complete Hypergraphs

Dhruv Mubayi^{*} Yi Zhao[†]

Department of Mathematics, Statistics, and Computer Science

University of Illinois at Chicago

851 S. Morgan Street, Chicago, IL 60607

June 7, 2005

Abstract

Let $n \geq t \geq 3$. Suppose that $\mathcal{F}$ is a family of t -element subsets of $[n]$ satisfying $|\mathcal{F}| > \binom{n}{t-1}^{t-1} + o(n^{t-1})$. Our main result implies that there exists a t -element subset S of $[n]$ such that $\{E \cap S : E \in \mathcal{F}\}$ contains all 2-element subsets of S , and this is asymptotically sharp. When $t = 3$, we prove the same result whenever $|\mathcal{F}| > \left\lfloor \frac{(n-1)^2}{4} \right\rfloor$, which is sharp for all $n \geq 3$. Our proofs use the Erdős-Rado Sunflower Lemma and recent results on hypergraph Turán numbers.

1. Introduction

Let $[n] = \{1, 2, \dots, n\}$. Given a set X , 2^X denotes the family of all subsets of X , and $\binom{X}{q} = \{A \subseteq X : |A| = q\}$. A *hypergraph* $\mathcal{H}$ on X is a family of subsets of X ; these subsets are called *edges* of $\mathcal{H}$ and X is the *vertex* set of $\mathcal{H}$. A hypergraph is called p -uniform or a p -graph if all its edges have size p .

Let G be a hypergraph on X and $S \subseteq X$. The *trace* or *link* of G on S , is

$$G|_S := \{E \cap S : E \in G\}.$$

^{*}Research partially supported by National Science Foundation Grant DMS-0400812, and an Alfred P. Sloan Research Fellowship. Email: mubayi@math.uic.edu

[†]Research partially supported by NSA Grant H98230-05-1-0079, and a Campus Research Board grant at University of Illinois at Chicago. Email: zhao@math.uic.edu

Note that $G|_S$ is a simple hypergraph, *i.e.*, we discard multiplicity when considering $G|_S$.

If there exists a set S such that $G|_S$ contains a copy of F as a subhypergraph, we say that G contains F as a trace, or F is a trace of G . In this case we write $G \rightarrow F$ ($G \not\rightarrow F$ otherwise). Let $L^p(n, F)$ ($L(n, F)$) denote the maximum number of edges in a p -uniform (not necessarily uniform) hypergraph on $[n]$ not containing F as a trace. Extremal problems on traces started from determining $L(n, 2^{[t]})$. Sauer [13], Perles-Shelah [14], and Vapnik-Chervonenkis [16] independently found that $L(n, 2^{[t]}) = \binom{n}{0} + \dots + \binom{n}{t-1}$. For the uniform case, Frankl and Pach [6] showed that $L^p(n, 2^{[t]}) \leq \binom{n}{t-1}$ for $t \leq p \leq n$. Many intersecting problems and applications on traces can be found in the survey of Füredi and Pach [8].

In this paper we consider the problem of forbidding a level of the lattice $2^{[t]}$ as a trace. More precisely, given integers p, t, n with $\max\{p, t\} \leq n$, we study the value of $L^p(n, \binom{[t]}{q})$ for $1 \leq q \leq t-1$ (the $q=0$ and $q=t$ cases are trivial). Frankl and Pach [6] studied the $q=1$ case and obtained that $\text{ex}(p+t-1, \binom{[t]}{1}) \leq L^p(n, \binom{[t]}{1}) \leq \binom{p+t-1}{t-1}$, where ex is the Turán number that we discuss later. Balogh, Keevash and Sudakov [1] investigated the trace problem of forbidding more than one non-trivial level of $2^{[t]}$.

The problem of determining $L^p(n, \binom{[t]}{q})$ becomes trivial when $p < q$, so we assume that $p \geq q$ and note that its $p=q$ case is the well-known Turán problem. We let $2 \leq q \leq t-1$ throughout the paper. We only consider the *uniform* trace number because Füredi and Quinn [9] showed that $L(n, \binom{[t]}{q}) = L(n, 2^{[t]})$ for every $0 \leq q \leq t$, in other words, forbidding a non-trivial level of the lattice $2^{[t]}$ is equivalent to forbidding the whole lattice in the non-uniform case. Following graph theory language, the forbidden configuration $\binom{[t]}{q}$ is a complete q -graph on t vertices, so we denote it by K_t^q , and write $K_t = K_t^2$.

Our first result, which is little more than an observation, determines the order of magnitude of $L^p(n, K_t^q)$.

Proposition 1.1. *Fix $2 \leq q < \min\{p, t\}$. Then $L^p(n, K_t^q) = \Theta(n^{\min\{p, t-1\}})$.*

Our second result, which is also easy to prove, relates $L^p(n, K_t^q)$ when $p < t$ to the classical Turán number. Given a family $\mathcal{F}$ of r -graphs, the *Turán number* $\text{ex}(n, \mathcal{F})$ of $\mathcal{F}$ is the maximum number of edges in an r -graph on n vertices containing no $F \in \mathcal{F}$ (see *e.g.*, Füredi [7] for a survey). When $\mathcal{F} = \{F\}$, we write $\text{ex}(n, F)$ instead of $\text{ex}(n, \{F\})$. A trace problem can be viewed as a Turán problem. For example, if we denote K_3 by $\{12, 23, 31\}$, then $L^3(n, K_3) = \text{ex}(n, \{F_1, F_2, F_3\})$, where $F_1 = \{124, 234, 134\}$, $F_2 = \{124, 234, 135\}$, and $F_3 = \{124, 235, 136\}$. In general, for any q -graph F and $q \leq p$, we have $L^p(n, F) =$

$\text{ex}(n, \mathcal{H}^p(F))$, where $\mathcal{H}^p(F)$ is the family of all p -graphs H with $|F|$ edges such that $H \rightarrow F$. Let $H_{q,t}^p$ be the member of $\mathcal{H}^p(K_t^q)$ with the maximum number of vertices. In other words, $H_{q,t}^p$ is the p -graph obtained from K_t^q by enlarging each of its $\binom{t}{q}$ edges with a (different) set of $p - q$ new vertices. Trivially $H_{p,t}^p = K_t^p$.

Proposition 1.2. *Fix $2 \leq q < p < t$. Then $L^p(n, K_t^q) = \text{ex}(n, H_{q,t}^p) + o(n^p)$.*

Our main result determines $L^p(n, K_t^q)$, $p \geq t$ in many cases.

Theorem 1.3. *Fix $2 \leq q < t \leq p$. When either $q = t - 1$ or $p = t$, we have*

$$L^p(n, K_t^q) = L^{t-1}(n, K_t^q) + o(n^{t-1}). \quad (1)$$

When $p \leq t$ or $q = t - 1$, Proposition 1.2 and Theorem 1.3 together imply that

$$L^p(n, K_t^q) = \text{ex}(n, H_{q,t}^r) + o(n^r). \quad (2)$$

where $r = \min\{p, t - 1\}$. This means that determining $L^p(n, K_t^q)$ asymptotically is as difficult as the hypergraph Turán problem. For example, (2) implies that $L^4(n, K_4^3) = \text{ex}(n, K_4^3) + o(n^3)$, and determining $\text{ex}(n, K_4^3)$ is a well-known open problem of Turán [15]. When $r = 2$, the range of parameters above forces $(q, t) = (2, 3)$. In this case, Theorem 1.3 and Mantel's Theorem yield that for every fixed $p \geq 3$,

$$L^p(n, K_3) = \frac{n^2}{4} + o(n^2). \quad (3)$$

Determining $\text{ex}(n, H_{q,t}^p)$ in general seems hopeless. The special case $q = 2$ was recently studied by the first author [11] and Pikhurko [12]. A p -graph is ℓ -partite if its vertices can be partitioned into ℓ classes, such that every edge has at most one vertex from each class (thus $p \leq \ell$). An ℓ -partite p -graph is called *complete* if it contains all allowable edges. We denote by $T_\ell^p(n)$ the complete ℓ -partite p -graph (a *generalized* Turán graph) on n vertices with no two class sizes differ more than one. Clearly $T_{t-1}^p(n)$ contains no $H_{2,t}^p$ as a subgraph and

$$|T_\ell^p(n)| = \sum_{S \in \binom{[\ell]}{p}} \prod_{i \in S} \left\lfloor \frac{n+i-1}{\ell} \right\rfloor = \binom{\ell}{p} \left(\frac{n}{\ell}\right)^p + o(n^p).$$

The first author [11] showed that $\text{ex}(n, H_{2,t}^p) = |T_{t-1}^p(n)| + o(n^p)$ as $n \rightarrow \infty$. Pikhurko [12] improved this to $\text{ex}(n, H_{2,t}^p) = |T_{t-1}^p(n)|$ for sufficiently large n . Moreover, it is easy to see that $T_{t-1}^p(n)$ not only contains no $H_{2,t}^p$ as a subgraph, but also contains no K_t^q for any $q \geq 2$

as a trace. In fact, every t -element vertex set S must contain two vertices a, b from the same vertex class. However, $T_{t-1}^p(n)$ has no edge containing both a and b . Consequently $T_{t-1}^p(n)|_S \not\supseteq K_t^q$ for any t -element set S . Together with the result of Pikhurko, this implies that for $2 < p < t$ and sufficiently large n ,

$$L^p(n, K_t) = |T_{t-1}^p(n)| = \text{ex}(n, H_{2,t}^p). \quad (4)$$

Combining (4) with the $(p, q) = (t, 2)$ case of Theorem 1.3 gives the result mentioned in the abstract (in fact, the asymptotic result in [11] suffices).

Corollary 1.4. *Fix $t \geq 3$. Then*

$$L^t(n, K_t) = |T_{t-1}^{t-1}(n)| + o(n^{t-1}) = \left(\frac{n}{t-1}\right)^{t-1} + o(n^{t-1}).$$

We believe that both Proposition 1.2 and Theorem 1.3 can be improved, at least, (1) should be true for all $p > t$. We conjecture the values of $L^p(n, K_t^q)$ as follows.

Conjecture 1.5. *Fix p, q, t, n with $2 \leq q < \min\{t, p\}$. For $n > n_0$, we have*

$$L^p(n, K_t^q) = \begin{cases} \text{ex}(n, H_{q,t}^p) & \text{if } p < t, \\ L^{t-1}(n-p+t-1, K_t^q) = \text{ex}(n-p+t-1, H_{q,t}^{t-1}) & \text{if } p \geq t. \end{cases}$$

The equation (4) confirms the conjecture for the case of $q = 2$, $p < t$, and sufficiently large n . As another evidence of the conjecture, we prove its smallest non-trivial case: $(p, q, t) = (3, 2, 3)$. Note that this sharpens (3) when $p = 3$.

Theorem 1.6. *Let $n \geq 3$. Then*

$$L^3(n, K_3) = \text{ex}(n-1, K_3) = \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

2. Preliminary Results

In this section we prove Proposition 1.1, Proposition 1.2 and the supersaturation property for trace numbers.

We have observed in the introduction that $T_{t-1}^p(n) \not\supseteq K_t^q$ whenever $2 \leq q \leq p < t$. The following proposition gives another way of constructing a hypergraph that does not contain K_t^q as a trace. Suppose that $p' < p$. Given a p' -graph G , we obtain a p -graph $\hat{H}^p(G)$ by adding a set C (called *core*) of $p - p'$ new vertices and replacing each $E \in G$ by $E \cup C$.

Proposition 2.1. Fix $2 \leq q \leq p' < p$ and $q < t$. If $G \subseteq \binom{[n]}{p'}$ satisfies $G \not\vdash K_t^q$, then $\hat{H}^p(G) \not\vdash K_t^q$. In particular, this implies that $L^p(n, K_t^q) \geq L^{p-1}(n-1, K_t^q)$.

Proof. The last assertion follows by letting G be the $(p-1)$ -graph on $[n-1]$ with the maximum number of edges containing no K_t^q as a trace. To prove the first claim, consider a t -element set E in $G' = \hat{H}^p(G)$. If E contains a vertex x from the core C , then all edges of G' contain x , and consequently all q -element subsets of $E \setminus \{x\}$ are absent from $G'|_E$. Otherwise $E \cap C = \emptyset$, and we have $G'|_E = G|_E \not\vdash K_t^q$. $\square$

Proof of Proposition 1.1. We need to show that $L^p(n, K_t^q) = \Theta(n^{\min\{p, t-1\}})$ for $2 \leq q < \min\{p, t\}$. When $p \geq t$, by the result of Frankl and Pach, $L^p(n, K_t^q) \leq \binom{n}{t-1}$. When $p < t$, trivially $L^p(n, K_t^q) \leq \binom{n}{p}$. We now consider lower bounds. When $p \leq t-1$, Since $T_{t-1}^p(n) \not\vdash K_t^q$ for $q \geq 2$, we have $L^p(n, K_t^q) \geq |T_{t-1}^p(n)| = \Omega(n^p)$. When $p \geq t$, let $G = \hat{H}^p(T_{t-1}^{t-1}(n-p+t-1))$. Since $T_{t-1}^{t-1}(n-p+t-1) \not\vdash K_t^q$, by Proposition 2.1, we have $G \not\vdash K_t^q$. This implies that $L^p(n, K_t^q) \geq |T_{t-1}^{t-1}(n-p+t-1)| = \Omega(n^{t-1})$. $\square$

Definition 2.2. Let F be a p -graph ($p \geq 2$) on $[\ell]$ and $\vec{m} = \langle m_1, \dots, m_\ell \rangle$ be a vector of positive integers. The blow-up $F(\vec{m})$ of F is obtained by replacing each vertex i by a vertex class V_i of size m_i , and each edge $\{i_1, \dots, i_p\}$ by the family of all p -element sets $\{w_1, \dots, w_p\}$, where $w_j \in V_{i_j}$. We simply write $F(m)$ if all $m_i = m$.

A phenomenon discovered by Brown, Erdős and Simonovits [4], usually called *supersaturation*, implies that $\text{ex}(n, F(\vec{m})) = \text{ex}(n, F) + o(n^r)$ for every r -graph F and its blow-up $F(\vec{m})$. To prove Proposition 1.2, we need a lemma from [11], which is a simple consequence of supersaturation.

Lemma 2.3 (Lemma 4 in [11]). Let m, p be positive integers with $p \geq 2$, and let $\mathcal{F}$ be a finite family of p -graphs. If H is a p -graph satisfying $H \subseteq F(m)$ for all $F \in \mathcal{F}$, then $\text{ex}(n, H) \leq \text{ex}(n, \mathcal{F}) + o(n^p)$. $\square$

Proof of Proposition 1.2. We need to show that $L^p(n, K_t^q) = \text{ex}(n, H_{q,t}^p) + o(n^p)$ for fixed $2 \leq q < p < t$ and $n \rightarrow \infty$. Recall that $L^p(n, K_t^q) = \text{ex}(n, \mathcal{H}^p(K_t^q))$, where $\mathcal{H}^p(K_t^q)$ consists of all p -graphs with $\binom{t}{q}$ edges containing K_t^q as a trace, including $H_{q,t}^p$. Thus $L^p(n, K_t^q) \leq \text{ex}(n, H_{q,t}^p)$. On the other hand, it is easy to see that $H_{q,t}^p \subseteq F(\binom{t}{q})$ for each $F \in \mathcal{H}^p(K_t^q)$. By Lemma 2.3,

$$\text{ex}(n, H_{q,t}^p) \leq \text{ex}(n, \mathcal{H}^p(K_t^q)) + o(n^p) = L^p(n, K_t^q) + o(n^p). \quad (5)$$

This gives the desired lower bound for $L^p(n, K_t^q)$. □

Next we prove the supersaturation phenomenon for trace problems.

Lemma 2.4. *Fix p, q, t, m with $q < p < t$. Then*

$$L^p(n, K_t^q(m)) \leq L^p(n, K_t^q) + o(n^p).$$

Proof. Recall that $\mathcal{H}^p(K_t^q(m))$ is the family of p -graphs whose $|K_t^q(m)|$ edges contain $K_t^q(m)$ as a trace, and $H_{q,t}^p$ is obtained from K_t^q by enlarging each of its $\binom{t}{q}$ edges with a different set of $p - q$ new vertices. Let $\tilde{H} = H_{q,t}^p(\vec{m})$, where $m_i = m$ for all the vertices v_i in the original K_t^q , and $m_i = 1$ for the other vertices. It is easy to see that $\tilde{H}$ is a member of $\mathcal{H}^p(K_t^q(m))$. We thus have

$$\begin{aligned} L^p(n, K_t^q(m)) &= \text{ex}(n, \mathcal{H}^p(K_t^q(m))) \\ &\leq \text{ex}(n, \tilde{H}) \\ &\leq \text{ex}(n, H_{q,t}^p) + o(n^p) \\ &\leq L^p(n, K_t^q) + o(n^p), \end{aligned}$$

where the first inequality holds because $\tilde{H} \in \mathcal{H}^p(K_t^q(m))$, the second inequality holds because of supersaturation for the Turán problems, and the last one holds because of (5). □

3. Proof of Theorem 1.3

The main idea in proving Theorem 1.3 is as follows: given a p -graph $G \not\vdash K_t^q$, we find a one-to-one function from G (or a subgraph containing most edges of G) to a p' -graph G' for some $q \leq p' < p$ such that $G' \not\vdash K_t^q$ (or $G' \not\vdash K_t^q(m)$). One way to find such a one-to-one function is by considering (co)degree. Let G be a hypergraph and S be a subset of its vertex set. The *degree* of S in G , $\deg_G(S)$, is the number of edges in G containing S (usually called *codegree* when $|S| \geq 2$). Given a p -graph G , if every edge $E \in G$ contains a p' -element subset E' with $\deg_G(E) = 1$, then $\phi(E) = E'$ defines a one-to-one function from G to $G' = \{E' : E \in G\}$.

Proposition 3.1. *Let $q < \min\{p, t\}$ and G be a p -graph such that $G \not\vdash K_t^q$. If each edge $E \in G$ contains a q -element subset E' with $\deg_G(E') = 1$, then $G' = \{E' : E \in G\}$ contains no K_t^q as a subgraph.*

Proof. Suppose instead, that G' contains a subgraph G_1 on a t -element set T such that $G_1 \cong K_t^q$. We will show that the trace of $\{E \in G : E' \in G'_1\}$ on T is G_1 , contradicting the assumption that $G \not\rightarrow K_t^q$. Let ϕ denote the function such that $\phi(E) = E'$ for every $E \in G$. Since ϕ is one-to-one, we can use ϕ^{-1} for its inverse function. We claim that each edge $E \in G$ with $\phi(E) \in G'_1$ satisfies that $E \cap T = \phi(E)$ and therefore $G|_T \supseteq K_t^q$. In fact, if $E \cap T \not\supseteq \phi(E)$, then $E \cap T$ contains another q -element set $E'' \in G'_1$. Since $E \neq \phi^{-1}(E'')$, the fact that both E'' and $\phi^{-1}(E'')$ contain E'' implies that $\deg_G(E'') \geq 2$, contradicting the definition of ϕ . $\square$

Proposition 3.2. *Let $2 \leq q \leq p' \leq p$, let G be a p -graph and $\phi : G \rightarrow \binom{[n]}{p'}$ be a function such that for each $E \in G$, $\phi(E)$ is a p' -element subset of E . If $G \not\rightarrow K_t^q$, then $\phi(G) \not\rightarrow K_t^q(m)$ with $m = \binom{t}{q}(p - q) + 1$.*

Proof. Suppose instead, that $\phi(G) \rightarrow K_t^q(m)$. This implies that there are disjoint vertex sets $X_1, X_2, \dots, X_t$ of size m such that for each q -element set Q having non-empty intersection with exactly q X_i 's, there exists a p -element set $P \in G$ such that $Q \subset P$. We call a t -tuple $T \in X_1 \times \dots \times X_t$ *bad* (otherwise *good*) if T contains some Q and at least one vertex from $P \setminus Q$. The number of bad t -tuples is at most $\binom{t}{q} m^q (p - q) m^{t-q-1}$. When $m > \binom{t}{q}(p - q)$, this number is less than m^t , the total number of t -tuples in $X_1 \times \dots \times X_t$. Hence there always exists a good t -tuple T and consequently $G|_T \supseteq K_t^q$, a contradiction. $\square$

The following lemma is the key observation for proving the $q = t - 1$ case of Theorem 1.3.

Lemma 3.3. *Fix $p \geq t \geq 2$, let S be a p -element set and $\mathcal{A}$ be a family of proper subsets of S . If every $(t - 1)$ -element subset of S is contained in some member of $\mathcal{A}$, then $\mathcal{A} \rightarrow K_t^{t-1}$.*

Proof. Induction on p . The base case $p = t$ is trivial, since every $(t - 1)$ -element subset of S is a member of $\mathcal{A}$, or $\binom{S}{p-1} \subseteq \mathcal{A}$. For the induction step, there are two cases. If $\binom{S}{p-1} \subseteq \mathcal{A}$, then fix a t -element set $T \subset S$, we have $\mathcal{A}|_T \supseteq \binom{T}{t-1}$ (since each $(t - 1)$ -element subset T' of T is contained in $T' \cup (S \setminus T)$, which is a member of $\mathcal{A}$). Otherwise $\binom{S}{p-1} \not\subseteq \mathcal{A}$, and there exists an $(p - 1)$ -element set $S' \notin \mathcal{A}$. It is easy to see that $\mathcal{A}|_{S'}$ satisfies the assumption of the lemma. We then apply the induction hypothesis to S' and $\mathcal{A}|_{S'}$ obtaining that $\mathcal{A}|_{S'} \rightarrow K_t^{t-1}$ and consequently $\mathcal{A} \rightarrow K_t^{t-1}$. $\square$

Finally we note the following fact: given a family $\mathcal{F}$ of p -graphs and an integer $c > 0$,

$$\text{ex}(n, \mathcal{F}) - \text{ex}(n - c, \mathcal{F}) = o(n^p), \quad \text{as } n \rightarrow \infty, \quad (6)$$

which immediately follows from the existence of $\lim_{n \rightarrow \infty} \text{ex}(n, \mathcal{F}) / \binom{n}{p}$.

Proof of Theorem 1.3 ($q = t - 1$). We need to prove that $L^p(n, K_t^{t-1}) = \text{ex}(n, K_t^{t-1}) + o(n^{t-1})$. Because of (6), it suffices to show that

$$\text{ex}(n - p + t - 1, K_t^{t-1}) \leq L^p(n, K_t^{t-1}) \leq \text{ex}(n, K_t^{t-1}).$$

The lower bound follows from $\hat{H}^p(G_0)$, where G_0 is an extremal graph of K_t^{t-1} on $n - p + t - 1$ vertices. To prove the upper bound, let G be an n -vertex p -graph not having K_t^{t-1} as a trace. Each edge $E \in G$ must contain a $(t - 1)$ -element subset E' with $\deg_G(E') = 1$, otherwise we may apply Lemma 3.3 to E and $G|_E - \{E\}$ to conclude that $G \rightarrow K_t^{t-1}$. We thus define $\phi(E) = E'$ and ϕ is a one-to-one function from G to $\binom{[n]}{t-1}$. By Proposition 3.1, the resulting $(t - 1)$ -graph G' contains no K_t^{t-1} as a subgraph, thus $|G| = |G'| \leq \text{ex}(n, K_t^{t-1})$. $\square$

The main step in proving the $p = t$ case of Theorem 1.3 is Lemma 3.5 below, which roughly says that if a p -graph G satisfies $G \not\rightarrow K_t^q$, then most edges in G can be reduced to a $(p - 1)$ -element set via a one-to-one function. The proof of Lemma 3.5 needs the Sunflower Lemma of Erdős and Rado [3]. A *sunflower* (or Δ -*system*) with k petals and a core Y is a collection of distinct sets $S_1, \dots, S_k$ such that $S_i \cap S_j = Y$ for all $i \neq j$.

Lemma 3.4 ([3]). *Let G be a p -graph with $|G| > p!(k - 1)^p$. Then G contains a sunflower with k petals.* $\square$

Lemma 3.5. *Let G be an n -vertex t -graph such that $G \not\rightarrow K_t^q$, and define $G' = \{E \in G : \deg_G(S) \geq 2 \text{ for each } S \in E \text{ with } |S| = t - 1\}$. Then $|G'| = O(n^{t-2})$ as $n \rightarrow \infty$.*

Proof. When $q = t - 1$, the definition of G' implies that for any $E \in G'$, each $(t - 1)$ -element subset of E is contained in a member of G different from E . This implies that $G|_E$ contains a copy of K_t^{t-1} . Hence $G' = \emptyset$ and we are done. We may thus assume that $q \leq t - 2$.

Since $G \not\rightarrow K_t^q$, each t -element set $E \in G$ contains a q -element subset Q , denoted by $\psi(E)$, such that

$$\tilde{E} \cap E \neq Q \quad \text{for every } \tilde{E} \in G. \tag{7}$$

In order to show that $|G'| = O(n^{t-2})$, it suffices to show that for each set $Q \in \binom{[n]}{q}$,

$$|G_Q| = O(n^{t-2-q}), \quad \text{where} \quad G_Q = \{E \in G' : \psi(E) = Q\}.$$

If $|G_Q| \leq t!(t-q)^t$, then we are done, since t, q are constants but n approaches ∞ . Otherwise we apply the Sunflower Lemma (Lemma 3.4) to G_Q and obtain a sunflower $E_1, \dots, E_{t-q+1}$ with a core C . The definition of G_Q implies that $Q \subseteq C$; on the other hand, clearly $|Q| \leq t-1$. We claim that $|C| \geq q+2$. In fact, if $C = Q$, then $E_1 \cap E_2 = Q$, contradicting (7). Now suppose that $C = Q \cup \{x\}$ for some vertex x . Let $Y = E_1 \setminus \{x\}$. By the definition of G' , $\deg_G(Y) \geq 2$, and consequently there exists a vertex y such that $Y \cup \{y\} \in G$. Since the number of petals is $t-q+1 \geq 3$, there exists a set $E_i, i \neq 1$ such that $y \notin E_i$. Then $(Y \cup \{y\}) \cap E_i = Q$, again contradicting (7).

We next observe that no $E \in G_Q$ satisfies $E \cap C = Q$. Otherwise, because the number of petals is $t-q+1 > |E \setminus C|$, there exists a set E_i from the sunflower such that $E_i \cap (E \setminus C) = \emptyset$. Thus $E_i \cap E = Q$, contradicting (7). Since a set in G_Q contains Q as a subset, we conclude that every $E \in G_Q$ satisfies $|E \cap C| \geq q+1$. Now we partition G_Q into $G^1 \cup G^2$, where G^1 consists of those $E \in G_Q$ with $|E \cap C| = q+1$.

It is easy to see that $|G^2| \leq 2^{|C|-q} \binom{n-|C|}{t-q-2} = O(n^{t-q-2})$. Now let us estimate $|G^1|$. Fix a $(q+1)$ -element subset T of C and let $G_T^1 = \{E \setminus T : E \in G^1, E \cap C = T\}$. We observe that G_T^1 contains no matching of size $t-q+1$, since such a matching would create a sunflower in G^1 with $t-q+1$ petals and a core of size $q+1$, which is impossible as we discussed earlier. A result in [5] (Page 1317, Proposition 11.6) says that every n -vertex r -graph not containing a matching of size s has at most $(s-1) \binom{n-1}{r-1}$ edges. We thus have $|G_T^1| \leq (t-q) \binom{n-|C|-1}{t-(q+1)-1} = O(n^{t-q-2})$. Consequently, $|G^1| \leq \binom{|C|}{q+1} O(n^{t-q-2}) = O(n^{t-q-2})$. This finally gives that $|G_Q| = |G^1| + |G^2| = O(n^{t-q-2})$. $\square$

Proof of Theorem 1.3 ($p = t$). We first show the lower bound that $L^t(n, K_t^q) \geq L^{t-1}(n, K_t^q) - o(n^{t-1})$. In fact, by Proposition 2.1, $L^t(n, K_t^q) \geq L^{t-1}(n-1, K_t^q)$; by (6) and the fact that $L^{t-1}(n, K_t^q)$ is a Turán number, $L^{t-1}(n-1, K_t^q) \geq L^{t-1}(n, K_t^q) - o(n^{t-1})$. Putting these together, we obtain the desired lower bound.

In order to prove $L^t(n, K_t^q) \leq L^{t-1}(n, K_t^q) + o(n^{t-1})$, we consider an n -vertex t -graph G with $G \not\rightarrow K_t^q$. We first apply Lemma 3.5 to G and conclude that $|G'| = O(n^{t-2})$ for G' defined in the lemma. Since each $E \in G \setminus G'$ contains a $(t-1)$ -element subset of degree 1 (in G), we obtain a one-to-one function $\phi : G \setminus G' \rightarrow \binom{[n]}{t-1}$. Next we apply Proposition 3.2 to $G \setminus G'$ and obtain that $\phi(G \setminus G') \not\rightarrow K_t^q(m)$ for some integer m . Hence $|G \setminus G'| = |\phi(G \setminus G')| \leq L^{t-1}(n, K_t^q(m))$. Applying Lemma 2.4 with $p = t-1$, we further

have $|G \setminus G'| \leq L^{t-1}(n, K_t^q) + o(n^{t-1})$. This finally implies that

$$|G| = |G \setminus G'| + |G'| \leq L^{t-1}(n, K_t^q) + o(n^{t-1}) + O(n^{t-2}) = L^{t-1}(n, K_t^q) + o(n^{t-1}). \quad \square$$

4. An Exact Result

In order to prove Theorem 1.6, we need the following lemma, whose can be proved by following the original proof of Mantel's Theorem. We use $+$ instead of $\cup$ for a disjoint union. In a graph G , given a vertex set A and a vertex x , $N(x, A)$ denotes the neighborhood of x in A , and $d(x, A) = |N(x, A)|$, in particular $d(x) = d(x, V(G))$. For disjoint vertex sets X and Y , we denote by $e(X, Y)$ the number of edges between X and Y . For simplicity we write ab instead of $\{a, b\}$.

Lemma 4.1. *Let $G = (V, E)$ be a triangle-free graph such that*

$$\text{for every } ab \in E, \text{ there exists } c \in V, \text{ such that } ac \notin E \text{ and } bc \notin E. \quad (\star)$$

Then $|E| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 1$ with equality only when G has the following structure: $V(G) = A + B + \{z\}$, there exist $a \in A$ and a non-empty set $B_z \subseteq B$ such that $E(G) = A \times B - \{ab : b \in B_z\} + \{zb : b \in B_z\} + \{az\}$.

Proof. Let xy be an edge. Since G is triangle-free, we have $N(x) \cap N(y) = \emptyset$. With $(\star)$, we further derive that $d(x) + d(y) \leq n - 1$.

If $d(x) + d(y) \leq n - 2$ for every edge xy in G , then following Mantel's proof of his theorem, we have

$$\frac{4|E|^2}{n} = \frac{(\sum_{x \in V} d(x))^2}{n} \leq \sum_{x \in V} (d(x))^2 = \sum_{xy \in E} (d(x) + d(y)) \leq (n - 2)|E|,$$

$$|E| \leq \frac{n(n-2)}{4} < \frac{(n-1)^2}{4} < \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 1.$$

Otherwise assume that $d(x) + d(y) = n - 1$ for some $e = \{x, y\}$. Let $A = N(y)$ and $B = N(x)$. We know that $A \cap B = \emptyset$ and $A \cup B = V - \{z\}$ for some vertex z . Let $d_1 = d(z, A)$ and $d_2 = d(z, B)$.

Case 1: $d_1 = 0$, or $d_2 = 0$.

Say, $d_1 = 0$. For each $b \in N(z, B)$, there exists $a \in A$ such that $ab \notin E$, since otherwise edge xb does not satisfy $(\star)$. This implies that

$$|E| = e(A, B) + d(z, B) \leq |A||B| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor.$$

Case 2: $d_1, d_2 > 0$.

In this case $d_1 d_2 - d_1 - d_2 + 1 = (d_1 - 1)(d_2 - 1) \geq 0$ with equality if and only if at least one of d_1, d_2 is 1. Since G is triangle-free, there is no edge between $N(z, A)$ and $N(z, B)$. Thus $e(A, B) \leq |A||B| - d_1 d_2$ and

$$|E| = e(A, B) + d(z, A) + d(z, B) \leq |A||B| - d_1 d_2 + d_1 + d_2 \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor + 1,$$

where equality holds only when G has the desired structure. $\square$

Proof of Theorem 1.6. To show that $L^3(n, K_3) \geq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor$, we enlarge each edge of $K_{\lfloor \frac{n-1}{2} \rfloor, \lceil \frac{n-1}{2} \rceil}$ with the same new vertex.

To prove the upper bound, we consider a 3-graph H on $[n]$ such that $H \not\rightarrow K_3$. The proof of the $q = t - 1$ case of Theorem 1.3 implies that each triple $T \in H$ contains a pair $\phi(T)$ with $\deg_H(\phi(T)) = 1$. We thus obtain a graph G on $[n]$ with edge set $E = \{\phi(T) : T \in H\}$. Clearly $|E| = |H|$, and G satisfies $(\star)$ because

$$\text{if a triple } \phi(\{a, b, c\}) = ab, \text{ then } ac \notin E \text{ and } bc \notin E. \quad (8)$$

Next we claim that $G \neq G^*$, where G^* is a graph causing the equality in Lemma 4.1. Suppose, to the contrary, that $G = G^*$. Let us consider edges za and zb for any $b \in B_z$. By (8), $\phi^{-1}(za) = \{z, a, x\}$ for some $x \in A \setminus \{a\}$, and $\phi^{-1}(zb) = \{z, b, y\}$ for some $y \in B \setminus B_z$. Since a is the unique vertex which is non-adjacent to both x and b , we have $\phi^{-1}(xb) = \{a, b, x\}$. The trace of $\{z, a, x\}, \{z, b, y\}, \{a, b, x\}$ on $\{z, a, b\}$ is a K_3 , contradicting $H \not\rightarrow K_3$.

Finally we apply Lemma 4.1 and obtain that $|H| = |E| \leq \left\lfloor \frac{(n-1)^2}{4} \right\rfloor$. $\square$

5. An Open Problem

A general uniform trace problem is to determine $L^p(n, F)$ for arbitrary p and F . Because of the close connection between trace problems and Turán problems, as seen in (2), it is hopeless to determine $L^p(n, F)$ in general.

Let us consider $L^3(n, F)$ when F is a graph. Fix $t = \chi(F)$. When $t \geq 4$, we have

$$L^3(n, F) = |T_{t-1}^3(n)| + o(n^3) = \binom{t-1}{3} \left(\frac{n}{t-1} \right)^3 + o(n^3).$$

In fact, the lower bound for $L^3(n, F)$ follows from $T_{t-1}^3(n) \not\rightarrow F$, where $T_{t-1}^3(n)$ is the generalized Turán graph defined in the introduction. The reason for $T_{t-1}^3(n) \not\rightarrow F$ is that when embedding F into a $(t-1)$ -partite graph, some partition set must contain both ends of an edge of F . The upper bound follows from (4) and Lemma 2.4. The same arguments actually show that $L^p(n, F) = |T_{t-1}^p(n)| + o(n^{t-1})$ for every F with $t = \chi(F) > p$.

Problem 5.1. *Determine the order of magnitude of $L^3(n, F)$ for every F with $\chi(F) \leq 3$.*

This seems no easier than determining the order of magnitude of the Turán numbers for bipartite graphs. We can derive an upper bound for $L^3(n, F)$ as follows. A result of Erdős [2] implies that $\text{ex}(n, K_3^3(m)) = O(n^{3-\frac{1}{m^2}})$. For a 3-graph H , it is clear that $K_3^3(m) \subseteq H$ implies that $H \rightarrow K_3(m-1)$. For each F with $\chi(F) \leq 3$, there exists m such that $F \subseteq K_3(m)$. Hence $L^3(n, F) \leq L^3(n, K_3(m)) \leq \text{ex}(n, K_3^3(m+1)) = O(n^{3-c})$, where $c = 1/(m+1)^2$. However, we do not have a matching lower bound. For example, we only know $L^3(n, K_3(2)) = \Omega(n^{5/2})$, in contrast to an upper bound $O(n^{26/9})$ derived by above arguments (we can improve it to $O(n^{11/4})$ by using some extra ideas). This lower bound can be seen from the 3-partite 3-graph with partition sets A, B, C of size n , and the edge set $\{e \cup v : v \in C, e \in G\}$, where G is a maximum C_4 -free bipartite graph on (A, B) with $\Omega(n^{3/2})$ edges.

References

- [1] J. Balogh, P. Keevash and B. Sudakov, Disjoint representability of sets and their complements, J. Combinatorial Theory Ser. B, to appear.
- [2] P. Erdős, On extremal problems of graphs and generalized graphs, Israel J. Math, 2, (1964) 183–190.
- [3] P. Erdős, R. Rado, Intersection theorems for systems of sets, J. London Math. Soc. 35 (1960) 85–90.
- [4] P. Erdős, M. Simonovits, Supersaturated graphs and hypergraphs. Combinatorica 3 (1983), no. 2, 181–192.

- [5] P. Frankl, Extremal set systems. Handbook of combinatorics, Vol. 1, 2, 1293–1329, Elsevier, Amsterdam, 1995.
- [6] P. Frankl, J. Pach, On disjointly representable sets, *Combinatorica* 4 (1984) 39–45.
- [7] Z. Füredi, Turán type problems. Surveys in combinatorics, 1991 (Guildford, 1991), 253–300, London Math. Soc. Lecture Note Ser., 166, Cambridge Univ. Press, Cambridge, 1991.
- [8] Z. Füredi, J. Pach, Traces of finite sets: extremal problems and geometric applications. Extremal problems for finite sets (Visegrád, 1991), 251–282, Bolyai Soc. Math. Stud., 3, János Bolyai Math. Soc., Budapest, 1994.
- [9] Z. Füredi, F. Quinn, Traces of finite sets. *Ars Combin.* 18 (1984), 195–200.
- [10] Gy Katona, T. Nemetz, M. Simonovits, On a problem of Turán in the theory of graphs. *Mat. Lapok* 15 (1964) 228–238.
- [11] D Mubayi, A hypergraph extension of Turán’s theorem, *J. Combinatorial Theory Ser. B*, to appear.
- [12] O. Pikhurko, Exact Computation of the Hypergraph Turan Function for Expanded Complete 2-Graphs, submitted.
- [13] N. Sauer, On the density of families of sets, *J. Combinatorial Theory Ser. A* 13 (1972), 145–147.
- [14] S. Shelah, A combinatorial problem; stability and order for models and theories in infinitary languages, *Pacific J. Math.* 41 (1972), 247–261.
- [15] P. Turán, On an extremal problem in graph theory. *Mat. Fiz. Lapok* 48 (1941), pp. 436–452.
- [16] V. N. Vapnik and A. Ya. Chervonenkis, On the uniform convergence of relative frequencies of events to their probabilities, *Theory Probab. Appl.* 16 (1971), 264–280.